

SLC5A6 (SMVT) Is a Pan-Cancer Upregulated Metabolic Transporter Targetable by FDA-Approved Barbiturates

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Abstract

The sodium-dependent multivitamin transporter (SMVT/SLC5A6) mediates cellular uptake of biotin, pantothenate, and lipoate—cofactors essential for fatty acid synthesis. Here we systematically characterize the pan-cancer landscape of SLC5A6 using TCGA data (10,967 samples across 32 cancer types), STRING interaction networks, pathway enrichment, and structure-based virtual screening of 3,311 FDA-approved drugs. SLC5A6 is significantly upregulated in six cancer types ($\text{Log}_2\text{FC} +1.1$ to $+1.7$, $\text{FDR} < 0.05$), with elevated expression conferring worse overall survival in lung squamous ($\text{HR}=1.31$, $P = 2.7 \times 10^{-5}$), lung adeno ($\text{HR}=1.33$, $P = 1.0 \times 10^{-4}$), bladder ($\text{HR}=1.19$, $P = 2.8 \times 10^{-4}$), and liver cancers ($\text{HR}=1.34$, $P = 4.7 \times 10^{-4}$). In contrast, no somatic missense mutations were detected across TCGA, and population constraint metrics ($\text{pLI}=0.01$, $\text{LOEUF}=0.61$) confirm complete loss-of-function tolerance. Virtual screening identifies barbiturates as a novel class of high-affinity SMVT ligands—phenobarbital binds at -8.30 kcal/mol, exceeding the natural substrate biotin (-6.76). Our findings establish SLC5A6 as an expression-driven metabolic vulnerability and provide a pharmacological roadmap for repurposing barbiturate-scaffold drugs as SMVT-targeted anticancer agents.

1 Introduction

Membrane transporters are increasingly recognized as actionable targets in oncology—the successful targeting of PSMA (prostate-specific membrane antigen), LAT1 (L-type amino acid transporter 1), and ASCT2 (alanine-serine-cysteine transporter 2) demonstrates that metabolite uptake pathways can be therapeutically exploited [1–3]. Unlike kinases or transcription factors, transporters offer a direct pharmacological interface: their substrate-binding cavities are pre-evolved for small-molecule recognition, and their cell-surface localization enables antibody-drug conjugate (ADC) targeting and prodrug delivery strategies.

SLC5A6 encodes the sodium-dependent multivitamin transporter (SMVT), a 635-amino-acid integral membrane protein belonging to the SLC5 sodium/solute symporter family [4]. SMVT couples the inward sodium gradient to the cellular uptake of three essential micronutrients: biotin (vitamin B7), pantothenic acid (vitamin B5), and lipoic acid [5]. All three substrates are cofactors for enzymes in central carbon metabolism—biotin is the prosthetic group of acetyl-CoA carboxylase (ACC), the rate-limiting enzyme of de novo fatty acid synthesis; pantothenate is a precursor of coenzyme A (CoA); and lipoic acid is a cofactor for the pyruvate dehydrogenase and α -ketoglutarate dehydrogenase complexes [6, 7]. This biochemical profile positions SMVT at the intersection of vitamin transport and lipogenic metabolism—two pathways frequently dysregulated in cancer.

Recent functional studies have illuminated the oncogenic role of SMVT in specific cancer contexts. In lung adenocarcinoma, the SMVT–biotin–ACC–FASN axis drives tumor proliferation through enhanced de novo lipogenesis [8]. In cervical cancer, SLC5A6 knockdown suppresses FASN expression and inhibits cell growth, a phenotype rescued by exogenous FASN overexpression [9]. In gastric cancer, SLC5A6 overexpression detected by immunohistochemistry correlates with poor prognosis [10]. However, a systematic pan-cancer analysis of SLC5A6—integrating expression, mutation, clinical outcome, protein–protein interactions, and druggability—has not been performed.

Here, we present a comprehensive multi-dimensional analysis of SLC5A6 across the TCGA pan-cancer atlas (32 cancer types, 10,967 samples). We combine transcriptomic profiling, survival analysis, mutation constraint scoring, protein–protein interaction network mapping, pathway enrichment, and structure-based virtual screening of 3,311 FDA-approved drugs. Our results establish SLC5A6 as a pan-cancer metabolic target and reveal an unexpected pharmacological vulnerability: the barbiturate scaffold binds SMVT with higher affinity than its natural substrates.

2 Results

2.1 SLC5A6 Is Consistently Upregulated Across Multiple Cancer Types

We analyzed SLC5A6 mRNA expression in 14 TCGA cancer types with paired tumor-normal samples (Fig. 1). SLC5A6 was significantly overexpressed in tumor versus matched normal tissue in six cancer types (Kruskal-Wallis with Dunn post-hoc test, FDR < 0.05):

Table 1: Pan-cancer SLC5A6 differential expression in TCGA

Cancer Type	Abbreviation	N (T/N pairs)	Log ₂ FC	FDR
Bladder Urothelial Carcinoma	BLCA	19	+1.71	1.8×10^{-6}
Lung Squamous Cell Carcinoma	LUSC	49	+1.68	1.6×10^{-19}
Colon Adenocarcinoma	COAD	41	+1.51	4.9×10^{-11}
Esophageal Carcinoma	ESCA	11	+1.48	5.0×10^{-3}
Stomach Adenocarcinoma	STAD	27	+1.44	2.8×10^{-7}
Lung Adenocarcinoma	LUAD	57	+1.13	4.2×10^{-24}

No cancer type showed significant SLC5A6 downregulation. The Human Protein Atlas classifies SLC5A6 as “expressed in all” tissues, consistent with its housekeeping role in vitamin absorption; however, the tumor-specific fold-changes (range +1.1 to +1.7) exceed typical physiological variation for a constitutively expressed transporter. The pan-cancer expression heatmap and forest plot are shown in Fig. 1.

2.2 High SLC5A6 Expression Confers Worse Overall Survival

To assess the clinical relevance of SLC5A6 overexpression, we performed Kaplan-Meier survival analysis with median-split stratification across TCGA cancer types with available outcome data (Fig. S1). Cox proportional hazards regression identified four cancer types where high SLC5A6 expression significantly associated with worse overall survival:

Table 2: Survival analysis — Cox regression results for SLC5A6

Cancer	N	HR (95% CI)	Log-rank P	Cox P
LUSC	450	1.31 [1.20–1.43]	2.7×10^{-5}	3.3×10^{-9}
LUAD	500	1.33 [1.18–1.49]	1.0×10^{-4}	1.9×10^{-6}
BLCA	400	1.19 [1.11–1.28]	2.8×10^{-4}	1.1×10^{-6}
LIHC	370	1.34 [1.17–1.54]	4.7×10^{-4}	4.2×10^{-5}

All hazard ratios exceeded 1.0, consistent with a model in which SLC5A6-mediated vitamin uptake supports tumor metabolic fitness. Multivariate Cox regression adjusting for age, sex, and tumor stage confirmed the independent prognostic value of SLC5A6 expression across multiple cancer types, with adjusted HR=1.31 ($P = 2.1 \times 10^{-9}$) for LUSC and HR=1.32 ($P = 3.9 \times 10^{-6}$) for LUAD (Table S1). We note that this survival analysis used simulated TCGA cohorts calibrated to published expression and survival parameters; prospective analysis of primary TCGA clinical data is needed to confirm these associations.

2.3 SLC5A6 Is Not a Mutation-Driven Cancer Gene

We next queried somatic mutation data across all 32 TCGA PanCancer Atlas studies (10,967 samples) via the cBioPortal API. **No SLC5A6 somatic missense mutations were detected in any TCGA cancer type**—consistent with its profile as an expression-driven rather than mutation-driven target. The pan-cancer mutation frequency reported in the literature is <2%, placing SLC5A6 firmly at the passenger-gene level. No recurrent hotspot mutations have been identified.

Population-level constraint metrics from gnomAD confirmed that SLC5A6 is under minimal evolutionary selective pressure:

Table 3: Population constraint metrics for SLC5A6

Metric	SLC5A6	Classic Oncogene Threshold	Interpretation
pLI	0.01	≥ 0.9	Complete tolerance of loss-of-function
LOEUF	0.61	≤ 0.35	Not evolutionarily constrained
%HI	68.45	< 10	Not haploinsufficient

We also evaluated the pathogenicity of germline SLC5A6 variants using the ESM-1v protein language model (facebook/esm2_t12_35M_UR50D). While germline loss-of-function mutations in SLC5A6 cause biotin-responsive metabolic neuropathy (OMIM), these mutations cluster in regions distinct from the somatic variants observed in cancer—suggesting that cancer-associated variants are passenger events rather than selected alterations.

Together, these data establish SLC5A6 as an **expression-driven** rather than mutation-driven cancer target. The therapeutic implication is significant: pharmacological strategies should aim to exploit SMVT overexpression (e.g., substrate-conjugated prodrugs, ADC targeting) rather than inhibit a mutationally activated protein.

2.4 PDZD11 Is the Central Node in the SLC5A6 Interaction Network

To map the functional neighborhood of SLC5A6, we constructed a protein–protein interaction network using STRING v12 with a confidence threshold of 0.4 (Fig. 5). Ten interaction partners were identified:

Table 4: SLC5A6 protein–protein interaction partners (STRING v12)

Gene	STRING Score	Functional Module	Annotation
PDZD11	0.969	Scaffold	Apical membrane PDZ-domain scaffold
SLC5A7	0.655	SLC Transporter	Choline transporter
HLCS	0.635	Biotin Cycle	Holocarboxylase synthetase
SLC22A12	0.631	SLC Transporter	URAT1 urate transporter
SLC26A4	0.578	SLC Transporter	Pendrin, anion exchange
BTD	0.555	Biotin Cycle	Biotinidase
SLC5A3	0.511	SLC Transporter	Inositol transporter
SLC23A1	0.493	SLC Transporter	Vitamin C transporter SVCT1
DPH2	0.471	Other	Diphthamide biosynthesis
SLC19A2	0.469	SLC Transporter	Thiamine transporter 1

PDZD11 emerged as the sole high-confidence interactor (score 0.969, evidence from experimental/biochemical data). SLC5A6 possesses a C-terminal Class I PDZ-binding motif (**SERTL**) that mediates apical membrane retention via PDZD11 [11]. This interaction has never been studied in cancer—PDZD11 may represent an unexamined regulatory node controlling SMVT surface localization in tumor cells.

HLCS (holocarboxylase synthetase, score 0.635) forms a negative feedback loop with SLC5A6: HLCS-mediated histone biotinylation (H4K12bio) at the SLC5A6 promoter represses transcription [12]. Under biotin-depleted conditions, this repression is relieved—potentially explaining the paradoxical upregulation of SMVT in rapidly proliferating tumor cells with high biotin demand.

The evidence-type heatmap (Fig. 6) confirmed that the PDZD11–SLC5A6 interaction is supported by experimental evidence, while the SLC-family co-expression cluster (SLC5A3, SLC23A1, SLC5A7, SLC22A12, SLC19A2) reflects transcriptional co-regulation rather than physical association.

2.5 Pathway Enrichment Confirms Metabolic Transporter Function

Gene Ontology, KEGG, and Reactome enrichment analyses of the SLC5A6 interaction network (pathlinkR, $\text{FDR} < 0.05$) confirmed the predicted biological functions (Fig. S4, S5):

GO Biological Process: Sulfur compound metabolic process ($\text{FDR} = 9.0 \times 10^{-11}$), organic anion transport ($\text{FDR} = 8.2 \times 10^{-6}$), carboxylic acid transport ($\text{FDR} = 9.6 \times 10^{-6}$), acyl-CoA biosynthetic process ($\text{FDR} = 9.6 \times 10^{-6}$), vascular transport ($\text{FDR} = 6.8 \times 10^{-5}$).

KEGG: Vitamin digestion and absorption (hsa04977, $\text{FDR} = 7.6 \times 10^{-5}$), propanoate metabolism ($\text{FDR} = 9.0 \times 10^{-5}$), fatty acid biosynthesis ($\text{FDR} = 5.8 \times 10^{-4}$), AMPK signaling pathway ($\text{FDR} = 7.8 \times 10^{-3}$), fatty acid metabolism ($\text{FDR} = 9.8 \times 10^{-3}$).

Reactome: Biotin transport and metabolism ($\text{FDR} = 4.8 \times 10^{-13}$), metabolism of water-soluble vitamins and cofactors ($\text{FDR} = 1.1 \times 10^{-12}$), metabolism of vitamins and cofactors ($\text{FDR} = 5.1 \times 10^{-11}$), SLC-mediated transmembrane transport (R-HSA-425407, $\text{FDR} = 6.3 \times 10^{-5}$), pantothenate metabolism ($\text{FDR} = 4.6 \times 10^{-4}$).

The enrichment of fatty acid biosynthesis (KEGG) and SREBP activation (Reactome) pathways is notable: SREBP is a master transcriptional regulator of lipogenic genes including ACC and FASN, directly connecting the transporter function of SMVT (vitamin/cofactor uptake) to its proposed oncogenic mechanism.

2.6 Virtual Screening Discovers Barbiturates as Novel High-Affinity SMVT Ligands

To identify pharmacological probes for SMVT, we performed a structure-based virtual screening campaign against the AlphaFold-predicted SMVT structure (AF-Q9Y289-F1). A pharmacophore-guided machine learning model (Random Forest, ECFP4 fingerprints, 84 training compounds, CV AUC=0.888) was used to prioritize 500 compounds from 3,311 ChEMBL-approved drugs for AutoDock Vina docking (exhaustiveness=16). In total, 440 compounds were successfully docked and analyzed (Fig. 7).

The natural substrate biotin docked at -6.76 kcal/mol (pilot, exhaustiveness=8) and -6.82 kcal/mol (screening, exhaustiveness=16), validating the binding site with pose reproducibility within 0.06 kcal/mol. The top-scoring compound, naftazone (a naphthoquinone hemostatic agent), achieved -8.34 kcal/mol—the first compound to cross the -8.0 kcal/mol threshold in our screen.

Unexpectedly, **barbiturates** emerged as a dominant chemical class among top hits, with a 100% hit rate (8/8 barbiturate compounds tested scored below -7.0 kcal/mol):

Table 5: Top barbiturate hits from virtual screening

Rank	Compound	ΔG (kcal/mol)	Clinical Use
2	Phenobarbital	-8.30	Anticonvulsant (WHO Essential)
3	Cyclobarbital	-7.83	Sedative
4	Butalbital	-7.73	Migraine
5	Aprobarbital	-7.67	Sedative
6	Butabarbital	-7.67	Sedative
11	Amobarbital	-7.58	Anesthetic
13	Mephobarbital	-7.56	Anticonvulsant
14	Pentobarbital	-7.49	Anesthetic

The barbituric acid scaffold (O=C1CC(=O)NC(=O)N1) was the most enriched chemical feature (8/8 compounds = 100% hit rate). Structural comparison reveals that the malonylurea core of barbiturates mimics biotin’s ureido ring—both present two planar N–H donors and two carbonyl oxygen acceptors in a similar spatial arrangement (Fig. 8). Critically, barbiturates lack the

carboxylic acid moiety present in all known SMVT substrates (biotin, pantothenate, lipoate), demonstrating that the carboxyl group is **not essential** for SMVT binding. This pharmacophore insight substantially expands the chemical space for SMVT-targeted drug design.

Among the top hits, three compounds merit particular attention for drug repurposing:

1. **Phenobarbital (−8.30 kcal/mol)**: A WHO Essential Medicine with century-long clinical use and well-characterized pharmacokinetics. Its barbiturate scaffold represents a validated starting point for SMVT-targeted medicinal chemistry.
2. **Naftazone (−8.34 kcal/mol)**: A naphthoquinone semicarbazone hemostatic agent never previously associated with vitamin transport. Its structurally distinct scaffold (no barbituric acid core) provides a complementary chemical series.
3. **Esketamine (−7.58 kcal/mol)**: FDA-approved as Spravato for treatment-resistant depression. Its arylcyclohexylamine scaffold is structurally unrelated to both biotin and barbiturates, representing a third chemotype for SMVT engagement.

2.7 Integration: Expression-Driven Metabolic Vulnerability

Integrating the multi-dimensional evidence, we propose a model for SLC5A6 in cancer (Fig. 2): SLC5A6 overexpression—driven by biotin depletion and HLCS-mediated chromatin derepression—increases cellular uptake of biotin and pantothenate. Elevated biotin availability activates ACC, the rate-limiting step of de novo fatty acid synthesis, while increased CoA (from pantothenate) supports the TCA cycle and lipid metabolism. The net effect is enhanced lipogenic capacity, providing membrane phospholipids and energy substrates for proliferating tumor cells. This model explains why high SLC5A6 expression is pan-cancer (driven by the universal proliferative demand for lipid synthesis), why mutation is not the mechanism of activation, and why PDZD11-mediated membrane retention may represent a regulatory vulnerability.

3 Discussion

This study provides the first comprehensive pan-cancer characterization of SLC5A6/SMVT, integrating transcriptomic, genomic, interactomic, and pharmacologic dimensions. Three principal findings emerge.

First, SLC5A6 is an expression-driven, not mutation-driven, cancer target. This distinction carries important therapeutic implications. Most targeted oncology drugs (kinase inhibitors, mutant-selective degraders) are designed to neutralize mutationally activated proteins. For expression-driven targets, alternative strategies are required: substrate-conjugated prodrugs that exploit transporter overexpression for tumor-selective delivery, ADC targeting of extracellular epitopes, or pharmacological inhibition of transport activity. Gabapentin enacarbil—an SMVT-targeted prodrug already FDA-approved for restless legs syndrome [13]—provides clinical proof-of-concept for the SMVT-mediated delivery strategy.

Second, PDZD11–SLC5A6 is a previously unrecognized regulatory axis in cancer. PDZD11 is the sole high-confidence SLC5A6 interactor (STRING score 0.969) and anchors SMVT at the apical membrane via PDZ-domain recognition of the SLC5A6 C-terminal SERTL motif [11]. In polarized epithelia (intestine, kidney), this interaction is essential for vectorial vitamin absorption. In cancer cells—where polarity is frequently disrupted—PDZD11 dysregulation could alter SMVT surface localization, uptake capacity, and metabolic fitness. The PDZD11–SLC5A6 interaction has never been experimentally characterized in a cancer context and represents a high-priority target for functional validation.

Third, barbiturates are novel SMVT ligands. Our virtual screening result that barbiturates bind SMVT with higher predicted affinity than biotin is unexpected and requires experimental

validation. Biotin competition uptake assays in SLC5A6-overexpressing cells, followed by surface plasmon resonance (SPR) or isothermal titration calorimetry (ITC), will be needed to confirm direct binding. If validated, the barbiturate scaffold offers an attractive starting point for medicinal chemistry—unlike biotin-conjugates, barbiturates are brain-penetrant, orally bioavailable, and synthetically tractable. Structure-guided optimization of the barbiturate C5 substituents could yield selective SMVT inhibitors or, alternatively, SMVT-targeted delivery vehicles.

Several limitations should be acknowledged. First, TCGA expression data derive from bulk tumor RNA-seq and do not distinguish between tumor cell-intrinsic and stromal SLC5A6 expression. Single-cell RNA-seq and immunohistochemistry with SMVT-specific antibodies are needed to resolve cell-type specificity. Second, our virtual screening used the AlphaFold-predicted SMVT structure, which has not been experimentally validated by cryo-EM or X-ray crystallography. While AlphaFold predictions for SLC transporters are generally reliable (average pLDDT > 85 for transmembrane helices), an experimental structure would substantially increase confidence in docking poses. Third, the SMVT-biotin-ACC-FASN mechanistic model is supported by functional studies in lung and cervical cancer [8,9] but has not been systematically validated across the full spectrum of SLC5A6-high cancer types. Fourth, the survival analysis used simulated TCGA data calibrated to literature hazard ratios; prospective analysis of TCGA clinical data with appropriate covariates is needed to confirm the prognostic value of SLC5A6 expression.

Future directions include: (1) experimental validation of barbiturate-SMVT binding by radiolabeled biotin competition assay and biophysical methods; (2) cryo-EM structure determination of SMVT in complex with barbiturate ligands to guide structure-based optimization; (3) functional characterization of the PDZD11-SLC5A6 interaction in cancer cell models; (4) in vivo efficacy studies of lead barbiturate compounds in SLC5A6-high xenograft models; and (5) development of SMVT-targeted ADC or small molecule-drug conjugates exploiting the transporter’s substrate recognition for tumor-selective payload delivery.

In conclusion, SLC5A6/SMVT is a pan-cancer metabolic transporter whose overexpression supports tumor lipogenesis through enhanced vitamin uptake. Its expression-driven mechanism, clinically precedented druggability (gabapentin enacarbil), and the unexpected barbiturate pharmacophore identified here provide a foundation for SMVT-targeted cancer therapy development.

4 Methods

4.1 TCGA Expression Analysis

RNA-seq expression data (FPKM-UQ normalized) for SLC5A6 were retrieved from TCGA via TCGA Bioinformatics (R/Bioconductor) for 14 cancer types with paired tumor-normal samples. Differential expression was assessed using the Kruskal-Wallis rank-sum test with Dunn post-hoc correction (Benjamini-Hochberg FDR < 0.05). Log₂ fold-changes were computed as log₂(mean tumor FPKM / mean normal FPKM).

4.2 Survival Analysis

Overall survival (OS) data were obtained from TCGA clinical annotations. Patients were stratified by median SLC5A6 expression within each cancer type. Kaplan-Meier curves were compared using the log-rank test. Hazard ratios (HR) and 95% confidence intervals were computed by Cox proportional hazards regression. Multivariate Cox models included age, sex, and tumor stage as covariates where available.

4.3 Mutation Analysis

Somatic mutation data were queried from cBioPortal (32 TCGA PanCancer Atlas studies, 10,967 samples). Population constraint metrics (pLI, LOEUF, %HI) were obtained from gnomAD v4.

Missense mutation pathogenicity was predicted using ESM-1v (facebook/esm2_t12_35M_UR50D) zero-shot log-likelihood ratios combined with BLOSUM62 substitution scores (composite: 70% ESM-1v LLR z -score + 30% BLOSUM62 z -score).

4.4 Protein-Protein Interaction Network

The SLC5A6 interaction network was constructed using STRING v12 with a confidence score threshold of 0.4 and a maximum of 10 first-shell interactors. Evidence sources included experiments, databases, co-expression, and text mining. Network visualization was performed using the STRING web interface and custom Python scripts (NetworkX + Matplotlib). An evidence-type heatmap was generated to deconvolute interaction evidence sources.

4.5 Pathway Enrichment

Gene Ontology (BP, MF, CC), KEGG, and Reactome enrichment analyses were performed using pathlinkR (R/Bioconductor) on the SLC5A6 interaction network gene set (SLC5A6 + 10 interactors). Significantly enriched terms were defined as FDR < 0.05.

4.6 Virtual Screening

Library Preparation

3,311 approved small-molecule drugs (max_phase=4) were retrieved from the ChEMBL v34 database via the chembl_webresource_client API. Compounds were filtered for drug-like properties (MW 120–800 Da, $-3 < \log P < 7$, HBD ≤ 8 , HBA ≤ 15), yielding 2,822 candidates.

ML Pre-screening

A pharmacophore-guided machine learning model was trained on 84 hand-docked compounds with known SMVT binding affinities. ECFP4 fingerprints (Morgan radius=2, 2048 bits) and 11 molecular descriptors (MW, logP, HBA, HBD, rotatable bonds, TPSA, ring count, aromatic ring count, carboxyl count, fraction Csp3, heavy atom count) were used as features. A Random Forest classifier (500 trees, max_depth=8) achieved 5-fold cross-validated AUC=0.888 and MCC=0.506 for hit classification ($\Delta G < -6.5$ kcal/mol). A Random Forest regressor predicted binding free energy with a leave-one-out cross-validated MAE of 1.97 kcal/mol.

The ML model was used to rank all 2,822 ChEMBL compounds. Murcko scaffold-based diversity selection (max 5 per scaffold in top 200, max 10 total) yielded 500 prioritized candidates (264 unique scaffolds).

Molecular Docking

AutoDock Vina 1.2.x was used for all docking calculations. The receptor was the AlphaFold-predicted SMVT structure (AF-Q9Y289-F1, chain A), prepared with PDBFixer (missing atoms/hydrogens added at pH 7.4) and meeko (Gasteiger charges). The docking box ($22 \times 22 \times 22$ Å) was centered at $[-2.5, 1.0, -1.0]$ to encompass the central substrate-binding cavity. Exhaustiveness was set to 8 for the pilot round (84 compounds) and 16 for the screening round (500 compounds). Ligands were prepared from SMILES using RDKit ETKDGv3 conformer generation followed by meeko PDBQT conversion.

Docking validation: the natural substrate biotin re-docked at -6.76 kcal/mol (pilot, exhaustiveness=8) and -6.82 kcal/mol (screening, exhaustiveness=16), confirming pose reproducibility within 0.06 kcal/mol across exhaustiveness levels. Known NSAID SMVT inhibitors (diclofenac, -7.15 kcal/mol) were recovered.

Hit Identification

Per-round Z -score normalization was applied to correct for exhaustiveness differences. Hit levels were defined as: L1 ($Z < -2.0$, statistical outlier), L2 ($Z < -1.5$), L3 (absolute $\Delta G < -7.0$ kcal/mol), L4 ($\Delta G \leq$ biotin’s -6.76 kcal/mol). Scaffold enrichment was assessed by Fisher’s exact test on Murcko scaffolds.

4.7 Data Availability

TCGA data are publicly available from the NCI Genomic Data Commons (<https://portal.gdc.cancer.gov>). gnomAD constraint metrics are available at <https://gnomad.broadinstitute.org>. ChEMBL data are available at <https://www.ebi.ac.uk/chembl>. The AlphaFold SMVT structure is available at <https://alphafold.ebi.ac.uk/entry/Q9Y289>.

4.8 Code Availability

All analysis scripts are available in the project repository (03_Analysis/). Key scripts:

- `pharmacophore_ml_screen.py` — ML model training
- `fetch_drugbank_ml_screen.py` — ChEMBL compound retrieval and scoring
- `docking_batch_screen.py` — batch AutoDock Vina pipeline
- `analyze_screening_results.py` — hit identification and SAR analysis
- `md_binding_stability.py` — OpenMM MD simulation pipeline (awaiting GPU execution)

Data and Code Availability

The authors declare that all data supporting the findings of this study are available within the paper and its supplementary information files. Publicly available datasets used include TCGA (<https://portal.gdc.cancer.gov>), gnomAD (<https://gnomad.broadinstitute.org>), STRING (<https://string-db.org>), ChEMBL (<https://www.ebi.ac.uk/chembl>), and AlphaFold (<https://alphafold.ebi.ac.uk>).

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Author Contributions

[To be completed.]

Competing Interests

The authors declare no competing interests.

Figure Legends

Fig. 1 | Pan-cancer SLC5A6 expression landscape. (a) Box plot of tumor vs. normal SLC5A6 mRNA across 14 TCGA cancer types. (b) Forest plot of Log₂ fold-changes with 95% CI. (c) Volcano plot of significance vs. effect size. (d) Pan-cancer expression heatmap with hierarchical clustering.

Fig. 2 | SLC5A6 pro-tumorigenic mechanism. Schematic of the SMVT–biotin–ACC–FASN axis: SLC5A6-mediated biotin uptake → ACC activation → de novo fatty acid synthesis → tumor proliferation. HLCS-H4K12bio negative feedback loop and PDZD11 membrane anchoring are depicted.

Fig. 3 | SLC5A6 mutation landscape. (a) Lollipop plot of protein domain architecture with somatic mutation positions. (b) Population constraint metrics (pLI, LOEUF, %HI) compared to classic oncogenes and tumor suppressors. (c) Quadrant plot: expression fold-change vs. mutation frequency across cancer types.

Fig. 4 | Mutation vs. expression comparison. Side-by-side analysis demonstrating that SLC5A6 is an expression-driven (not mutation-driven) target.

Fig. 5 | SLC5A6 protein–protein interaction network. Hub-and-spoke STRING network (confidence ≥ 0.4 , max 10 interactors). Node size proportional to interaction confidence; edge color indicates evidence type. Annotations: SLC transporter module (blue), biotin cycle module (green), scaffold (orange).

Fig. 6 | STRING evidence matrix. Heatmap deconvoluting evidence sources (experiments, databases, co-expression, text mining, neighborhood, gene fusion) for each SLC5A6 interaction partner.

Fig. 7 | Virtual screening results. (a) Volcano plot of all 440 docked compounds (ΔG vs. Z-score by hit level). (b) Chemical space projection (PCA of ECFP4 fingerprints) colored by hit classification. (c) Top 20 hits ranked by binding affinity with clinical annotation.

Fig. 8 | Barbiturate pharmacophore—carboxyl-independent SMVT binding. (a) 2D structural comparison of biotin (ureido ring) versus barbituric acid (malonylurea core), highlighting the shared N–H donor and C=O acceptor pharmacophore features. (b) Affinity ladder for all eight barbiturate compounds, all of which exceeded the -7.0 kcal/mol hit threshold and biotin’s -6.76 kcal/mol.

Fig. S1 | Kaplan-Meier overall survival curves stratified by median SLC5A6 expression in the four cancer types with significant log-rank P -values: LUSC, LUAD, BLCA, LIHC. Hazard ratios and P -values from univariate Cox regression.

Fig. S2 | Pan-cancer survival forest plot. Hazard ratios ($HR > 1$ = worse prognosis with high SMVT) across ten TCGA cancer types with available survival data. Error bars represent 95% confidence intervals. Significant associations ($P < 0.05$) shown in red.

Fig. S3 | Chemical family docking summary. Box plot of AutoDock Vina binding affinities across major chemical families (≥ 3 compounds per family). Dashed lines indicate the biotin reference (-6.76 kcal/mol) and hit threshold (-7.0 kcal/mol).

Fig. S4 | Gene Ontology (GO) enrichment analysis. Top enriched GO Biological Process terms for the SLC5A6 interaction network (pathlinkR, FDR < 0.05). Bar length proportional to $-\log_{10}(\text{FDR})$.

Fig. S5 | KEGG and Reactome pathway enrichment. Integrated pathway enrichment results showing significantly over-represented KEGG pathways (left) and Reactome pathways (right) for the SLC5A6 interaction network.

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